VINAY RAJ V

AI & Machine Learning Engineer

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LinkedIn: linkedin.com/in/vinay-raj-v | **GitHub:** github.com/vinayrajv2005 | **Portfolio:** my-portfolio-nmj4.vercel.app

✓ UK Registered Design Holder — Design #6497347 | Granted: 21 Jan 2026

Scan to view portfolio & projects

PROFESSIONAL SUMMARY

AIML undergraduate specializing in real-time AI systems using NLP, computer vision, and LLMs (RAG, Transformers). Experienced in building low-latency, production-ready AI applications with Flask/FastAPI and deep learning integration. Actively seeking AI/ML internships and research opportunities.

EDUCATION

B.E. in AI & Machine Learning JNNCE, Shivamogga | 2022–2026 | CGPA: 8.66	**Pre-University (12th Grade)** ACHARYA PU College, Shivamogga | 2020–2022 | 84%

TECHNICAL SKILLS

Languages	Python, Java, C
AI / ML	PyTorch, TensorFlow, Scikit-learn, HuggingFace, Whisper
NLP & CV	RAG, LangChain, FAISS, OCR (Tesseract), OpenCV, MediaPipe
Backend	Flask, FastAPI, Node.js, Express.js, REST APIs, WebRTC, Socket.IO
Tools	Pandas, NumPy, Power BI, Tableau, Git, GitHub, Streamlit, Colab

FEATURED PROJECTS

NEURON — RAG Document Chatbot 2026
Python, FastAPI, FAISS, LangChain, HuggingFace (Flan-T5)
- Offline multi-document RAG chatbot; FAISS semantic search ~35% better retrieval
- Multi-turn memory with source citations; zero external API dependency

ClassBridge AI — Smart Classroom 2026
Python, Flask, WebRTC, Whisper, HuggingFace, Socket.IO
- Real-time translation across 20+ languages with AI note-generation (~40% efficiency gain)
- WebRTC communication layer achieving <100ms latency

ProteinAI — Structure & Mutation Lab 2026
Python, PyTorch, BiLSTM, MLP, BLOSUM62, Three.js
- BiLSTM model ~90% accuracy for protein secondary structure prediction
- Interactive 3D molecular viewer with MLP mutation classifier using BLOSUM62

Fitness Guru — AI Personal Trainer 2025
Python, Flask, React, MediaPipe, OpenCV, Chart.js
- Real-time pose estimation (~95% accuracy) and injury detection (30% bad posture reduction)
- Analytics dashboard for workout history and progress tracking

LexSimplify — Legal NLP Platform 2026
Python, Flask, T5 Transformer, Tesseract OCR, HuggingFace
- T5-powered legal simplifier with risk-based clause detection across 15+ types
- Multilingual translation for 15+ languages incl. Kannada & Hindi; PDF/DOCX export

INTELLECTUAL PROPERTY

AI Based Water Quality Checking Device **✓ Granted** | Design #6497347 | UK IPO | Registered: 06 Jan 2026 | Granted: 21 Jan 2026
Class 10 · Subclass 05 | Instruments, Apparatus & Devices for Checking, Security or Testing
Inventors: Shaziya Banu Sharfudden, Dr. Pallavi Gundlupet Balakrishna, Dr. Ashwini Janagal Padmanabha, Khushan Gowda Halesh, **Vinay Raj Venkatesha**, Mohammed Ibrahim, Jeevan Hatturmatha Girish

ACHIEVEMENTS & CERTIFICATIONS

GenAI for Everyone — Coursera / Starweaver (Mar 2025)	MERN Stack & Full Stack Dev — Infosys Springboard (Jan 2025)
AI & ML Bootcamp — IIT Bombay Mood Indigo / Corizo (Jul 2025)	Artificial Intelligence: Types of AI — Infosys Springboard (Feb 2025)
Implementing AI with Amazon ML — Infosys Springboard (Mar 2025)	Python Data Structures — Great Learning Academy (Dec 2024)
SHE Secure 2025 National Hackathon Participant — JNNCE / Hack2Skill	